10-Minute Presentation Script

Use with Kritva_K_EL01_Final_Review_Presentation.pptx; pacing target totals 10:00

Delivery goal: explain the reasoning chain, not simply read numbers. Be candid about what is calculated versus assumed or still awaiting supplier confirmation. Keep the final slide brief enough to leave the review team ready for questions.

Slide	Target time	Speaking script / emphasis
1 — Title	0:30	“I am presenting K-EL/01, a split-yoke elbow actuator concept for the supplied robotic arm challenge. I focused on making the torque calculation auditable, separating torque transmission from radial support, and calling out what would need confirmation before any hardware release.”
2 — Inputs	1:00	“I took the masses and COM offsets directly from the brief: 1 kg at 350 mm, and 1.5 kg of forearm and hand at 180 mm. The initial configuration is horizontal, so this is a gravity-only static case. Output speed is 45 RPM; BLDC/PMSM, encoder, and planetary-preferred transmission are the stated architecture constraints. The link lengths shown are illustrative, not additional requirements. The brief does not define motion profile, accuracy, impacts or power-loss hold.”
3 — Calculation	1:30	“Using T equals m g r, payload contributes 3.4335 N·m; forearm and hand contribute 2.6487 N·m. Sum equals 6.0822 N·m. I used a preliminary factor of two, giving 12.1644 N·m target. This is an engineering assumption, not a certified safety factor. At 45 RPM, angular velocity is 4.712 rad/s and ideal output power at that target is 57.3 W before losses.”
4 — Drive	1:40	“The candidate is maxon EC-i 52 at 48 V, part 579165, with GP 52 C 91:1 candidate part 223094. Ratio times output speed gives 4095 RPM, only 375 RPM below the motor’s listed nominal speed. That is kinematically plausible but leaves limited headroom. At assumed 40% reducer efficiency, the simplified motor-side screen is 0.334 N·m, 143 W and about 3.64 A using torque constant. At 30%, it fails the simple continuous screen. The motor listing says 180 W class while nominal torque and speed multiply to about 198 W; I will ask the manufacturer for the authoritative rating point and loaded curve. Maxon also notes that EC-i iron-winding motors have a speed-dependent, nonlinear speed/torque characteristic, so I will not use the published gradient as a constant straight-line slope.”

Slide	Target time	Speaking script / emphasis
5 — Load path	1:40	“Torque runs from motor, through planetary gearbox and a candidate KTR ROTEX GS flexible coupling, to an independent output spindle, then flange and forearm. Radial loading is carried by the spindle and two spaced 6201 support bearings into the split-yoke and upper arm. That matters because the reducer’s torque rating alone does not establish a cantilever radial-load rating. The CAD bearing centres are 50 mm apart, and the forearm load plane is 50 mm beyond the rear support centre. The STEP model is a simplified envelope assembly, not production geometry.”
6 — Structural / error	1:20	“The two assigned masses give a simplified transverse force of 24.525 N. With a 50 mm overhang, the moment is 1.22625 N·m, explicitly converted to 1226.25 N·mm for the millimetre-based stress equation. A nominal Ø12 mm smooth shaft gives 7.23 MPa bending, 35.85 MPa torsional shear and about 62.52 MPa von Mises. Notch, fatigue, impact and deflection are excluded. One degree of average no-load backlash is about 6.1 mm at the 350 mm payload radius, so precision is an important trade-off.”
7 — Risks / next evidence	1:20	“The first major risk is dynamic and thermal performance. I would obtain the actual trajectory and duty, calculate peak and RMS torque, and get the configured supplier curve and efficiency. The second is output stiffness and load path: verify coupling hubs, spindle geometry, bearing life, fits and carrier stiffness. I would also clarify whether gravity must be held after power loss; there is no brake in this concept, so de-energized testing must be externally supported unless a rated hold solution is later designed.”
8 — Conclusion	1:00	“The concept package supplies the calculation, shortlist, CAD-equivalent STEP and source, labelled concept drawing, BOM, report, traceability, QA and a future validation plan. The assignment does not require buying or building hardware or detailed FEA. FreeCAD was not installed here, so the macro is supplied to generate a native file in FreeCAD. I am not claiming a supplier-confirmed stack, physical test, safety certification or manufacturing release. Thank you; I am ready to discuss the assumptions and closure plan.”

Timing note: the eight slide allocations total 10:00. Rehearse with a timer and use the detailed speaking cues rather than reading every word verbatim. Do not accelerate through the calculation or key supplier caveat. If interrupted, preserve the torque math, load path, open gates and concept-only boundary.